ENGINE CONTROL UNIT · KIRLOSKAR 3GK550ETA 4SR1

# Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

**24** blocks  all checks **passing**  generated from **sim/blocks/**  ← Field handbook

Sensor reference →

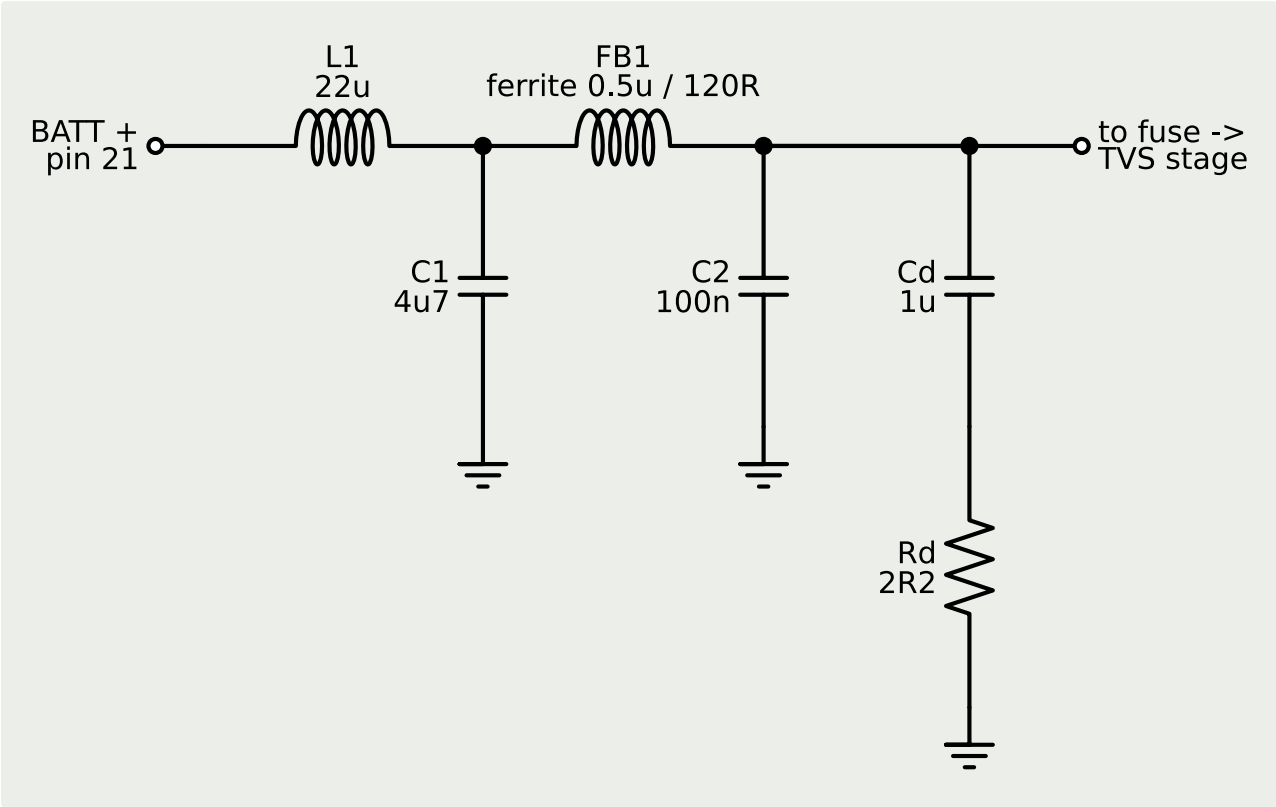

01 Power chain
02 Sensor front-ends
03 Speed and position
04 Injection
05 Actuators
06 Communications


## 01 Power chain

From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.

## EMI input filter `pin 21`

`18/18 checks`




Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

**Why it looks like that.** One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either — without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

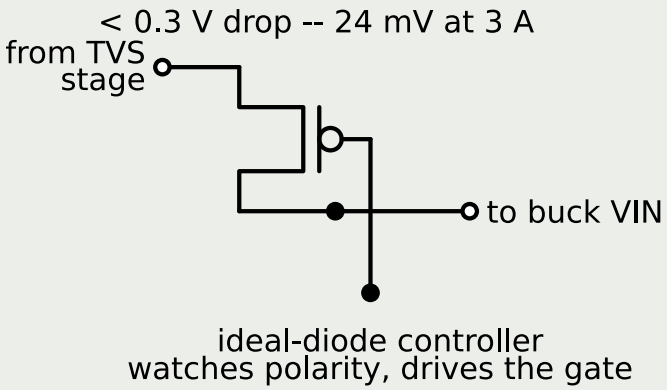
> **What the model does not cover.** Differential mode only. Common-mode noise — both wires swinging together against chassis, which is most real automotive failures — needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

| | | |
|---|---|---|
| PASS insertion loss at 150 kHz (band start) | | -40.564 dB |
| PASS insertion loss at 400 kHz (switching f) | | -50.734 dB |
| PASS insertion loss at 1 MHz | | -72.103 dB |
| PASS insertion loss at 30 MHz | | -129.540 dB |
| PASS insertion loss at 100 MHz (FM band) | | -95.679 dB |
| PASS worst insertion loss anywhere above 150 kHz | | -40.663 dB |
| PASS ... worst point sits at | | 151.5 kHz |
| PASS is 40 dB enough for Class 5? | unknown -- needs measured unfiltered emissions on hardware | |
| PASS ... IL@150kHz, C1 at X7R's own -15% temp derate (C1=3.74uF) | | -39.242 dB |
| PASS ... IL@400kHz, same corner | | -49.291 dB |
| PASS pin 21 DC current, upper bound (buck at its peak limit) | | 2.949 A |
| PASS L101 7447709220: saturation margin over the bound (Isat 6.5 A typ) | | 2.204 x |
| PASS ... self-heating at the bound, on a 105 C board | | 117.384 C |
| PASS ... under its 125 C operating maximum | | 117.384 C |
| PASS L102 744314047: saturation margin (Isat 9 A typ) | | 3.052 x |

| | |
|---|---|
| PASS F101 0437005.WRA: continuous rating at 105 C (80% x 79%) | 3.160 A |
| PASS ... over the pin 21 bound by | 1.072 x |
| PASS ... the largest 437A rating still rated 63 V (7 A is 35 V, under the 40 V load dump) | 63.000 V |

Netlist: sim/blocks/emi_filter.cir

## Reverse-battery protection – 10/10 checks




A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

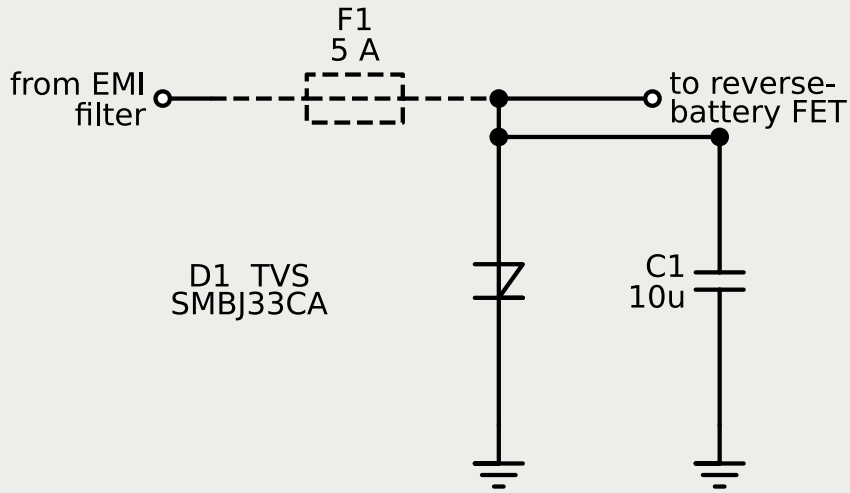
**Why it looks like that.** Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

> **What the model does not cover.** Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

| | | |
|---|---|---|
| PASS Schottky drop at 3 A | | 0.465 V |
| PASS ideal-diode FET drop at 3 A | | 0.024 V |
| PASS FET drop stays under 0.3 V spec to 5 A | | 0.040 V |
| PASS ... Schottky does NOT meet it | 0.52 V at 5 A -- fails the 0.3 V spec | |
| PASS ... at the sheet's specified ceiling, 40 mOhm at Vgs = -4.5 V | | 0.200 V at 5 A |
| PASS ... still inside the 0.3 V requirement by | | 1.500 x |
| PASS power burnt in the Schottky at 3 A | | 1.395 W |
| PASS ... and in the FET instead | | 0.072 W |
| PASS Schottky blocks -14 V (leak into 1k) | | 1.000 mV |
| PASS ideal-diode FET blocks -14 V | | 0.014 mV |

Netlist: sim/blocks/reverse_battery.cir

## Transient clamp `pin 21` `30/30 checks`




F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

**Why it looks like that.** Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

> **What the model does not cover.** It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load_dump, which did check this for its own pulse.

| | |
|---|---|
| PASS pulse actually applied (source peak) | |
| PASS clamped voltage at buck input | |
| PASS ... margin inside LM5164's 100 V rating | |
| PASS ... 60 V-class parts are NO LONGER viable downstream | |
| PASS ... so the reverse-battery FET needs 100 V class, not 60 | |
| PASS ... EVERY part on VBAT_PROT must clear this, not just the FET | |
| PASS ... so a 40 V-abs-max integrated driver is EXCLUDED from this rail | |
| PASS ... and the load dump alone already leaves it no margin | |
| PASS ... minimum class for anything battery-connected | 80 V or better: above the 73.3 V pulse 2a clamp with margin. 60 V |
| PASS rail table: LM5164 buck, VIN -- VIN abs max | |
| PASS rail table: Q1 reverse FET, P-ch -- \|Vds\| class | |

```
PASS rail
table: Q2
negative-clamp
FET -- Vds
class
```

```
PASS rail
table: D103
negative-clamp
Schottky -- Vr
class
```

```
PASS rail
table:
TPS40210 boost
ctrl, VDD
(220R + 36 V
clamp, 46.6 V
worst) --
clamped VDD
```

```
PASS rail
table: Boost
inductor / FET
(150 V class)
-- Vds class
```

```
PASS rail
table:
Injector
battery-side
blocking
diodes -- Vr
class
```

```
PASS rail
table: Relay
flyback diodes
(discrete_io)
-- Vr class
```

```
PASS rail
table: Relay
low-side FETs
(discrete_io)
-- Vds class
```

```
PASS rail
table:
Metering FET
and freewheel
(metering_egr)
-- Vds / Vr
class
```

```
PASS rail
table:
Injector low-
side FETs
(drain on
boost rail) --
Vds class
```

```
PASS rail
table: EGR
bridge FETs,
all four
(metering_egr)
-- Vds class
```

```
PASS ... what
the two
corrected
parts WERE
```

```
PASS GATE_KILL
unclamped vs a
20 V Vgs
rating (the
finding)
```

```
PASS GATE_KILL
with BZX84-
C12-Q (12.7 V
max at 5 mA)
-- margin
```

```
PASS ... hot:
BZX84-C12-Q
+10 mV/K max,
100 K above 25
C
```

```
PASS ... zener
current at the
pulse peak
```

```
PASS ... 47k
pullup
dissipation at
the pulse peak
```

```
PASS recovers
to nominal
after pulse
```

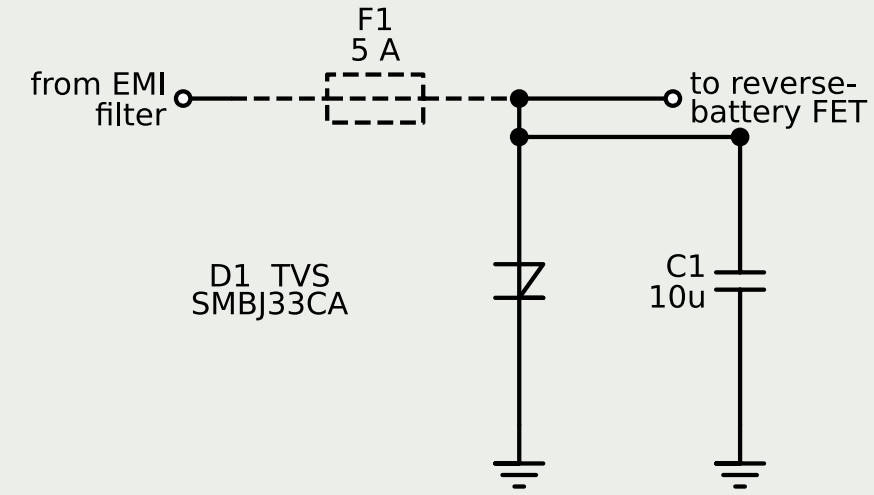
```
PASS F101:
pulse 2a I2t
through the
fuse
```

```
PASS ... as a
fraction of
0437005.WRA's
2.30 A2s
melting I2t
```

Netlist: sim/blocks/transient_clamp.cir

## Load dump pin 21

11/11 checks




pulse 5b: Us*=40 V, 0.5R, 400 ms
D1 absorbs ~46 J at ~116 W -- FAILS an SMBJ-class part's rating
F1 dashed (drawn, not in this netlist) does not rescue it either:
fault current is 3.04 A, 0.6x F1's rating, and won't clear in 400 ms

The same TVS and capacitor as transient_clamp, against ISO 7637-2 pulse 5b: a suppressed load dump, 40 V through 0.5 ohms for 400 ms — four orders of magnitude longer than pulse 2a.

**Why it looks like that.** Clamps to 38 V, comfortably inside every voltage rating in the chain — the same story as pulse 2a. But 400 ms is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly 46 J at ~116 W average, which is on the order of 100x an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at Us* = 35 V, the figure usually quoted for a 12 V suppressed dump, and the TVS never reaches its 36.7 V breakdown — 0.000 A, 0 J, no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated 100 V, rides 40 V without complaint. Energy absorption only becomes unavoidable under pulse 5a.

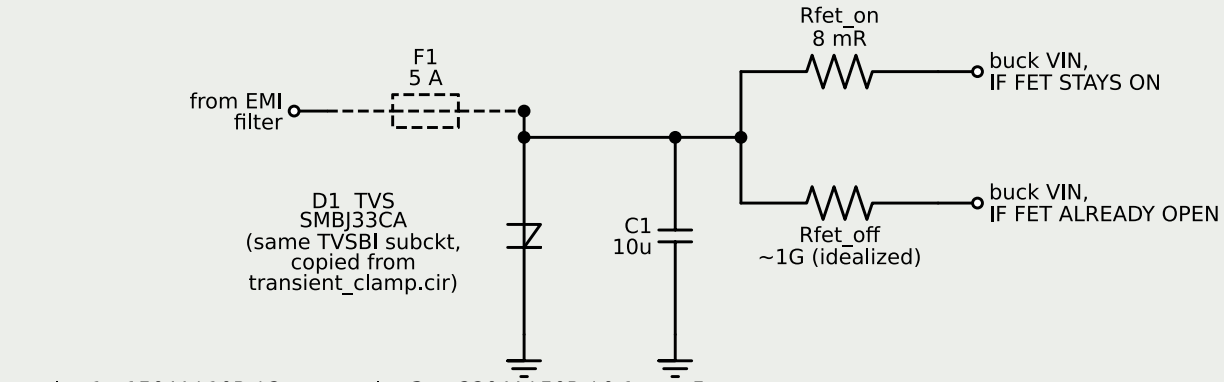
> **What the model does not cover.** Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

| | |
|---|---|
| PASS pulse actually applied (source peak) | 40.000 V |
| PASS clamped voltage through the 400 ms plateau | 39.604 V |
| PASS ... inside LM5164's 100 V rating | 2.525 x |
| PASS ... 60 V-class parts still viable | 39.604 V |
| PASS TVS current through the plateau -- now ZERO, the TVS stays off | 0.000 A |
| PASS average power dissipated in the TVS | 0.000 W |
| PASS total energy absorbed by the TVS over the pulse | 0.000 J |
| PASS energy stays within a single-pulse (~0.5 J) rating | 0.000 J |
| PASS average power stays within continuous (~3 W) rating | 0.000 W |
| PASS ... fraction of the continuous rating now used | 0.000 x |
| PASS recovers to nominal after pulse | 13.366 V |

Netlist: `sim/blocks/load_dump.cir`

## Negative transients `pin 21`

41/41 checks




The same TVS and bulk capacitor as transient_clamp/load_dump, against ISO 7637-2's two worst NEGATIVE pulses: pulse 1 (−150 V through 10 Ω for 2 ms, a supply disconnection from an inductive load) and pulse 3a (−220 V through 50 Ω for 0.1 ms, simulated as a 5-pulse burst at the standard's own 90 ms minimum repeat). Both Level IV, both sourced from research memo 06's own severity table. This became answerable only once the TVS stopped being modelled as a single diode, which conducts forward — and silently wrong — in reverse polarity.

**Why it looks like that.** The buck's own datasheet (LM5164, Absolute Maximum Ratings, fetched directly) rates VIN at −0.3 V to 100 V — the negative limit is not a mirror of the positive one. With the reverse-battery FET assumed to still be conducting — the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide — the buck's VIN pin sees −40.7 V on pulse 1 and −27.8 V on pulse 3a: 90–135× past that −0.3 V rating, not a thin margin. The reverse-battery stage does nothing to prevent this on the timescale these pulses occupy; it would help enormously if the controller reacts fast enough, and nothing here — or in reverse_battery.cir — says whether it does. Pulse 1's TVS energy (0.707 J single event) exceeds load_dump's own ~0.5 J class bound by 41%, a second genuine failure, not a manufactured one — TOLERANCE shows it flips to a PASS at Level III instead of IV, the same one-inequality shape load_dump's own Us* finding has. Pulse 3a's bulk capacitor (C1) does something transient_clamp never needed to consider: its own RC (500 µs) is five times the pulse's 90 µs flat top, so C1 arrests the excursion before the TVS ever reaches its 36.7 V breakdown at nominal value — the TVS

carries zero current, and nothing accumulates across the 5-pulse burst. TOLERANCE shows this is fragile: the same MLCC DC-bias derating emi_filter.cir already documents (up to 60% loss) is enough to flip the TVS into conducting at −38.4 V instead of −27.8 V — still a small energy (4.2 mJ/pulse, far under any rating), but a different regime, and the buck sees more negative voltage either way.

> **What the model does not cover.** The FET's actual turn-off delay is not modelled anywhere in this repo — reverse_battery.cir's own RESULT NOTE says its switch model 'will always look perfect in reverse' because it has none, and no controller part is chosen to pull one from a datasheet. Both bookends (FET fully on, FET already open) are simulated directly rather than guessing at a number in between. The burst is 5 repeats, not the standard's ~40000 (1 hr at 90 ms minimum repeat); nothing here rules out a slow thermal drift that would only show up over the full count, and TVS junction temperature is not modelled at all, the same gap load_dump's own RESULT NOTE states for its much larger energy question. Pulse 1's real ISO waveform is a decaying transient, not the flat top used here — pessimistic in the right direction, same simplification load_dump uses for pulse 5b.

| |
|---|
| PASS pulse 1 actually applied (source peak) |
| PASS pulse 3a actually applied (source peak) |
| PASS battery node, pulse 1 -- now set by the clamp diode, not the TVS |
| PASS buck VIN pin, pulse 1, WITH the negative clamp fitted |
| PASS ... evidence: a Schottky ALONE still misses LM5164's own -0.3 V abs max (sourced, not INFERRED), by |
| PASS battery node, pulse 3a -- now set by the clamp diode, not the TVS |
| PASS buck VIN pin, pulse 3a, WITH the negative clamp fitted |
| PASS ... evidence: same, pulse 3a -- a Schottky alone misses by |
| PASS RESOLVED: buck VIN, pulse 1, with the active clamp |
| PASS ... margin inside the rating |
| PASS ... the dip past the -100 mV engage threshold is slew x delay |
| PASS RESOLVED: buck VIN, pulse 3a, with the active clamp |
| PASS ... margin inside the rating |
| PASS ... and the improvement over the Schottky alone |
| PASS pulse 1: clamped VIN RECOVERS after the pulse |

| | |
|---|---|
| PASS ... it does not latch into a short on the returning source | |
| PASS pulse 3a: clamped VIN recovers after the fifth burst | |
| PASS drawn network: ENGAGE threshold at VIN | |
| PASS drawn network: RELEASE threshold at VIN | |
| PASS ... with TLV3201-Q1's full 5 mV offset against it, still below zero -- the clamp cannot hold on against reverse current | |
| PASS ... and the model's ACLAMP thresholds are these two numbers | |
| PASS ... the question this block carried since 20 Sep: is the sub-microsecond window before the clamp engages acceptable? | YES, and not marginally: VIN's fall rate is set by the 10 o |
| PASS ... comparison only: IF the FET had already opened (idealized, reverse_battery.cir's own optimistic model, not verified against any real turn-off delay) | |
| PASS ... same comparison, pulse 3a | |
| PASS ... so the reverse-battery stage helps only if it reacts in time, and nothing sources that time | FET-on branch tracks the ba |
| PASS bat1 recovers to its own Rsrc1/Rload divider baseline (11.25 V, not 13.5 V) | |

```
PASS bat3a recovers
to its own divider
baseline (6.75 V)
between every burst
pulse
```

```
PASS burst: peak-to-
peak spread across
repeats 2-5
(accumulation would
show up here)
```

```
PASS ... and repeat 1
matches them: nothing
is still settling at
5 ms
```

```
PASS ... no monotonic
drift across repeats
2-5
```

```
PASS TVS energy,
pulse 1 -- zero now
the clamp diode
conducts first
```

```
PASS ... so the CLAMP
DIODE now carries it:
peak current, pulse 1
```

```
PASS ... vs the 20 A
class already
specified for it
```

```
PASS ... clamp diode
energy, pulse 1
```

```
PASS ... peak clamp-
diode current, pulse
3a (50 ohm source, so
milder)
```

```
PASS ... which is the
flat-top current,
(206.5 - 0.4) / 50
ohm
```

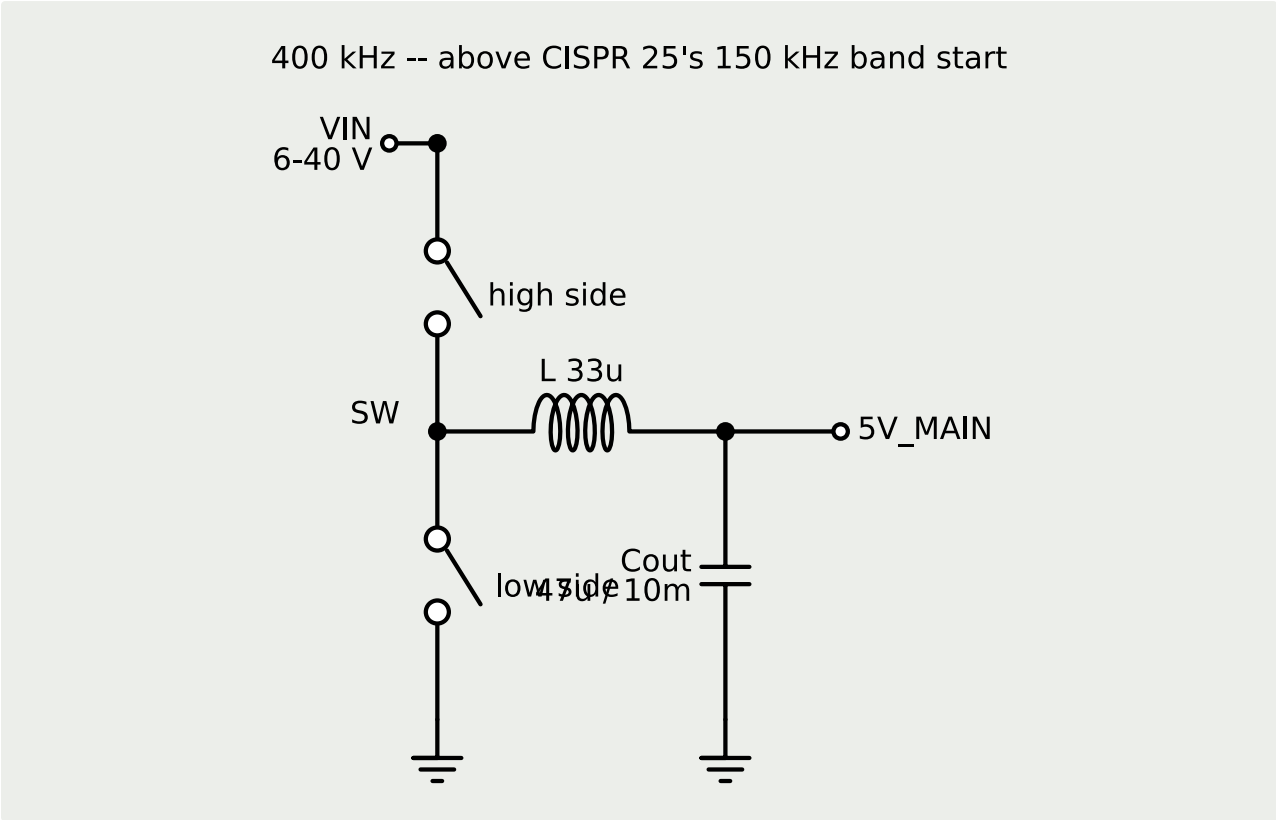
```
PASS ... vs
load_dump's own ~0.5
J single-pulse class
bound (the TVS now
clears it by not
conducting, not by
absorbing less)
```

```
PASS TVS energy,
pulse 3a (nominal C1
-- TVS stays below
its own breakdown,
does not conduct)
```

| | |
|---|---|
| PASS ... so nothing here can accumulate at nominal C1 -- see TOLERANCE for the corner where it would | TVS current is numerically zero thro |
| PASS F101: pulse 1 I2t through the fuse | |
| PASS ... as a fraction of 0437005.WRA's 2.30 A2s melting I2t (repeated pulse: 20% rule of thumb, pulse-cycling data open) | |

Netlist: sim/blocks/negative_pulses.cir

## Buck pre-regulator – 10/10 checks




6–40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

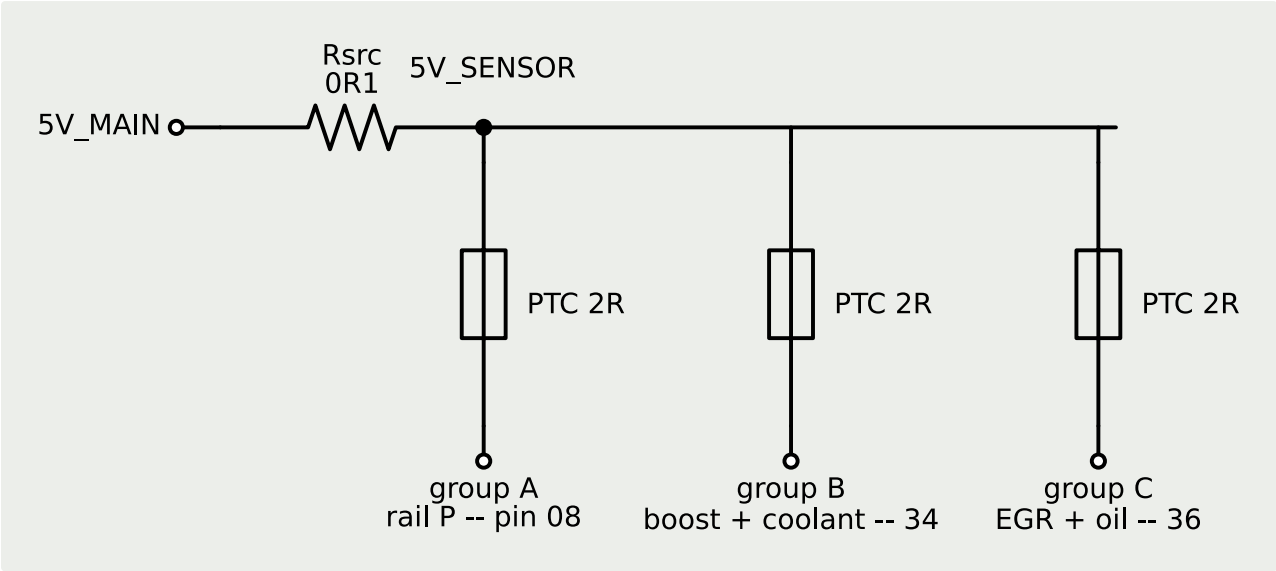
**Why it looks like that.** 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low — ripple current grows with input voltage even though duty cycle falls.

> **What the model does not cover.** Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

| | |
|---|---|
| PASS output at 13.5 V in (open loop, D=0.370) | 4.926 V |
| PASS output at 40 V in (open loop, D=0.125) | 4.945 V |
| PASS inductor ripple at 13.5 V | 0.256 A |
| PASS inductor ripple at 40 V | 0.348 A |
| PASS ... worst case is HIGH line, not low | 1.362 x |
| PASS worst-case ripple as fraction of 1 A load | 34.840 % |
| PASS output ripple at 13.5 V | 13.845 mV |
| PASS output ripple at 40 V | 21.423 mV |
| PASS L1 (7447714330) Isat hot against the 1.18 A worst peak | 1.720 x |
| PASS ... DCR loss at 1 A | 0.079 W |

Netlist: sim/blocks/buck_preregulator.cir

## 5V_SENSOR distribution – 22/22 checks




One tracking regulator per sensor group, behind one overvoltage cut-off, with pulse 2a applied through the real pin-21 filter.

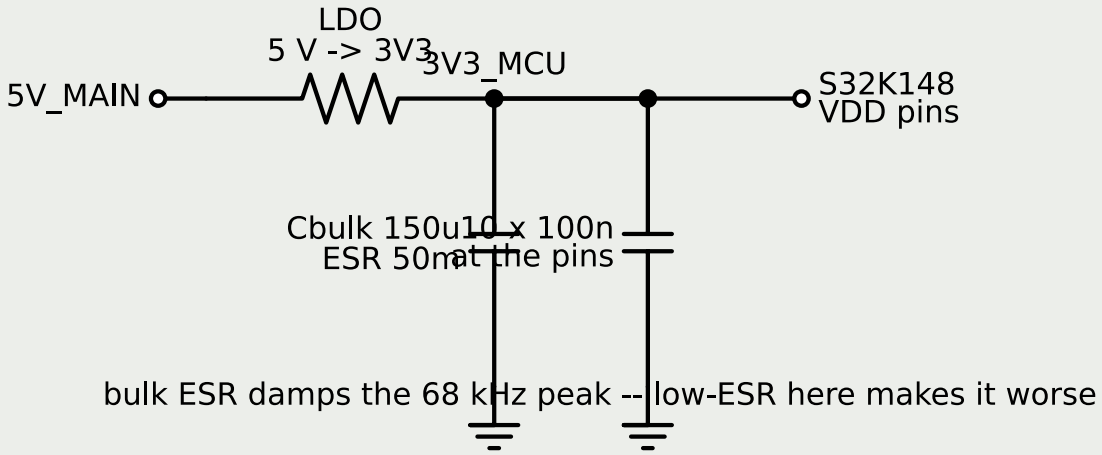
**Why it looks like that.** A shorted sensor harness is a routine fault. Each group's TPS7B4253-Q1 limits its own short at 301–520 mA, so the other two groups keep tracking 5V_MAIN and the MCU never sees the fault. The PTCs this replaced could not do that against a 1 A buck, and a battery short back-fed 5V_MAIN through them. The trackers survive 45 V and the rail reaches 73 V on pulse 2a, so a TPS48110-Q1 cut-off opens above 40.5–42.1 V and stays closed through the 40 V load dump. Its own inrush network (22 nF from gate to ground) also makes the FET a source follower for fast edges, so the trackers' input peaks near 22 V either way. The cut-off was chosen over an LTC4368 because it is AEC-Q100: decisions log section 7.

> **What the model does not cover.** The controller is modelled by function: a divider, a hysteretic comparator, a 2.6–4 µs delay, and a square-law FET with the datasheet's gate network. The trackers are datasheet arithmetic, not a transistor model. A sensor wire shorted to battery during pulse 2a is a double fault that this design does not cover; it is recorded.

| | |
|---|---|
| PASS pulse 2a, worst corner: trackers' input peak (VSNS) | 22.338 V |
| PASS ... against their 45 V absolute maximum | 22.338 V |
| PASS ... nominal corner | 22.073 V |
| PASS ... the OV comparator did open the FET (rail above VSNS) | 80.110 us |
| PASS all trackers at their limit, hold-up aged to 33 uF: VSNS low | 9.226 V |
| PASS ... headroom over 5 V + 0.52 V dropout | 3.706 V |
| PASS TPS48110-Q1 VS/CS slew through 100 ohm + 22 nF, top corner | 4.864 V/us |
| PASS TPS48110-Q1 VS/CS slew through 100 ohm + 22 nF, droop corner | 4.861 V/us |
| PASS clamp node with L1 NOT saturated (transient_clamp says 73.3) | 76.487 V |
| PASS ... still inside the 100 V class every part there is | 76.487 V |
| PASS OV trip, lowest tolerance | 40.521 V |
| PASS ... above the 40 V suppressed load dump by | 0.521 V |
| PASS OV trip, highest tolerance | 42.082 V |
| PASS load dump passes through: 40 V on the trackers' 45 V IN | 5.000 V |
| PASS inrush into 47 uF at turn-on (SLUSEE5E Eq. 3) | 0.162 A |
| PASS cranking at 6 V: headroom over 5V_MAIN + dropout | 0.460 V |
| PASS shorted group: its tracker limits at (max) | 0.520 A |
| PASS ... the other two still track 5V_MAIN to | 4.000 mV |
| PASS ... load on 5V_MAIN from three ADJ pins | 16.500 uA |
| PASS ... shorted tracker's dissipation at 13.5 V (thermal shutdown cycles it; the datasheet calls the output short-circuit proof) | 7.020 W |
| PASS output shorted to battery: 40 V load dump on OUT's -1..45 V | 40.000 V |
| PASS ... coincident with pulse 2a is a double fault -- recorded, not covered | decisions log section 7 |

Netlist: sim/blocks/sensor_rail.cir

## 3V3_MCU decoupling – 13/13 checks




The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

**Why it looks like that.** The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz — the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position. It was pulled a second time on 18 September, from 47 uF to 150 uF: at 47 uF the block cleared its target only at the nominal value of the LDO's output inductance, a number no chosen part pins down, and a 26% margin that evaporates when an unmeasured parameter is restated is not a margin. At 150 uF that resonance has dropped and damped enough that a different peak is now the tallest — the bulk/ceramic anti-resonance at 2.6 MHz, floored near 60 mOhm — and that one does not move with the LDO at all. The verdict has stopped depending on the unpinned number. More bulk beyond 150 uF buys nothing, because it does not touch the peak that now dominates.

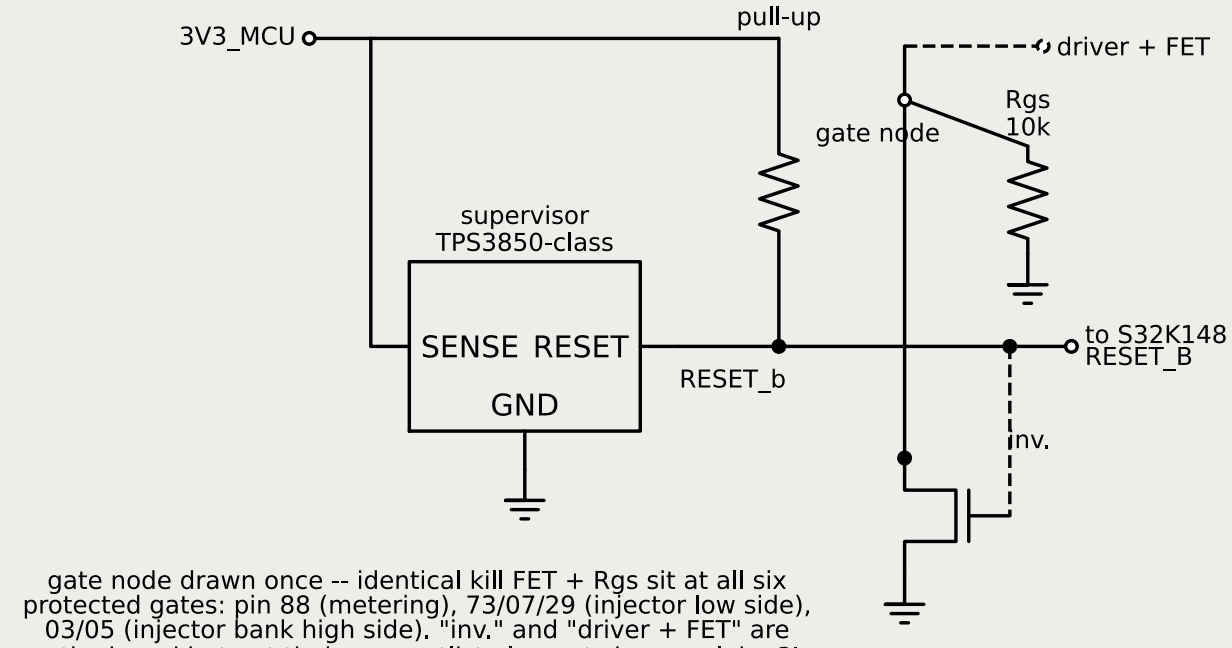
> **What the model does not cover.** Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz — which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

| | | |
|---|---|---|
| PASS peak impedance, 1 kHz to 100 MHz | | 60.355 mOhm |
| PASS ... and it has moved to the MHz anti-resonance | | 2.630 MHz |
| PASS ... i.e. bulk against ceramic, no longer regulator L against bulk | 2.63 MHz -- the LDO resonance is no longer the limiter | |
| PASS ten 100 nF parts at 100 MHz | | 46.923 mOhm |
| PASS one 1 uF part at 100 MHz | | 357.421 mOhm |
| PASS ... same capacitance, this much worse | | 7.617 x |
| PASS ten-part net at the 3.5 MHz anti-resonance | | 55.862 mOhm |
| PASS ... one-part net is LOWER there (its ESR damps it) | 34 mOhm vs 56 -- more ESR, better damping | |
| PASS ... evidence: peak with a WET electrolytic at its cold-ESR corner (Rb=200 mOhm, 4x) -- the corner that forced the spec | | 196.700 mOhm |
| PASS ... so a wet bulk part misses the 100 mOhm target by | | 96.700 mOhm |
| PASS RESOLVED: peak with the specified polymer bulk at its cold-ESR limit (Rb=50 mOhm) | | 60.355 mOhm |
| PASS ... and at the low-ESR end the same part reaches warm (Rb=20 mOhm) -- the under-damped side | | 92.446 mOhm |
| PASS ... worst of the two ends, margin under target | | 7.554 mOhm |

Netlist: sim/blocks/mcu_pdn.cir

## Supervisor / fail-safe kill path –

34/34 checks




gate node drawn once -- identical kill FET + Rgs sit at all six protected gates: pin 88 (metering), 73/07/29 (injector low side), 03/05 (injector bank high side). "inv." and "driver + FET" are on the board but not their own netlist elements here -- claim 3's FAIL (see RESULT NOTE) is about Rgs, drawn and modelled as 10k.

A TPS3850-class windowed supervisor watching 3V3_MCU — the same rail mcu_pdn.cir characterizes — whose RESET drives, through one inverting stage, a kill FET at every protected driver gate (pin 88 metering, 73/07/29 injector low side, 03/05 injector bank high side), in parallel with a standing 10 kΩ gate-source pulldown.

**Why it looks like that.** Research memo 11 found this board has no external watchdog, no brownout supervisor, and nothing defining output state during MCU reset — and that spec §3's other lines of defence both collapsed on 18 Sep 2026 (the fitted DSE4522 takes engine speed from this ECU over CAN, and its fuel relay signals rather than removes power), making this the first line, not defence-in-depth. Five falsifiable claims came out of that memo; two mattered most. Claim 4, the central one, PASSES with a wide margin: across six brownout ramps spanning 293 µs to 1 s, the kill path always finishes (gate below Vgs(th), fought by an adversarial driver the whole time) before 3V3_MCU crosses 2.97 V — the S32K148's own PLL-guarantee floor — with margin that GROWS at slower ramps, so the memo's flagged-open 'slow sag' case does not threaten it. Claim 3 FAILS, and not narrowly: swept across the memo's own illustrative Crss range under this design's own 2.8 V/µs boost-rail turn-off edge, the 10 kΩ pulldown alone lets the coupled gate voltage exceed a representative Vgs(th) starting at Crss = 50 pF — the LOW end of that range, not an extended one — and the memo's own hand arithmetic claiming '~5.6 mV' at 200 pF is off by three orders of

magnitude (560 µA through 10 kΩ is 5.6 V). A 300 Ω–1 kΩ pulldown clears it with real margin; the fix is a component VALUE, not a new part.

> **What the model does not cover.** The pin-mux gap is real and not papered over: which physical PTxx pins 88/73/07/29/03/05 land on is still a Phase-2 item, so their individual reset pull-state defaults cannot be stated — this model treats every protected gate generically, with no GPIO drive assumed at all except as a deliberately adversarial stimulus. The supervisor IC's own comparator propagation delay (assert side) is not sourced anywhere and is modelled as zero; claim 4's tightest margin (18 µs, at the fastest ramp) is the one that delay would eat into first. The inverting stage and kill FET's own gate-drive dynamics are collapsed into one switch with an INFERRED Ron. Ciss=2 nF and Vgs(th)=1.0 V are class figures, not a chosen part's datasheet numbers — no injector/metering FET has been selected yet.

| |
|---|
| PASS claim 1: gate below Vgsth, kill engaged (bound: 10-40 us) |
| PASS ... actual kill-path discharge time |
| PASS claim 1b: gate stays below Vgsth despite an adversarial driver actively holding it high (GPIO-independence) |
| PASS claim 2: final V(gate) with the 470 ohm pulldown fitted |
| PASS ... vs the same driver with no pulldown |
| PASS ... driver output impedance needed to still reach 9.5 V |
| PASS ... the 100 ohm driver this block assumed is too weak by |
| PASS claim 3 (evidence): peak V(gate), Crss=20 pF, Rpd=10k |
| PASS claim 3 (evidence): peak V(gate), Crss=50 pF, Rpd=10k |

| |
|---|
| PASS claim 3 (evidence): peak V(gate), Crss=100 pF, Rpd=10k |
| PASS claim 3 (evidence): peak V(gate), Crss=200 pF, Rpd=10k |
| PASS claim 3 (evidence): peak V(gate), Crss=500 pF, Rpd=10k |
| PASS claim 3 (evidence): peak V(gate), Crss=1000 pF, Rpd=10k |
| PASS ... memo 11's own hand arithmetic at Crss=200 pF claimed ~5.6 mV -- off by 1000x (560 uA * 10 kOhm = 5.6 V, not 5.6 mV); thi |
| PASS ... so the 10k pulldown fails even at the memo's OWN low end |
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=293 us |
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=585 us |
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1.17 ms |
| |

| |
|---|
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=10 ms |
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=100 ms |
| PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1000 ms |
| PASS ... tightest margin is at 18 us at 293 us -- the FASTEST ramp tried |
| PASS claim 5: kill-path latency vs one 26.67 ms cylinder interval |
| PASS ... caveat: excludes the watchdog's not own fault-DETECTION time |
| PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=220 ohm |
| PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=470 ohm |
| PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=1000 ohm |
| |

PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=2200 ohm

PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=4700 ohm

PASS ... same sweep read at Vgsth=2.5 V, the permissive end

PASS ... largest pulldown that holds the gate below Vgsth

PASS ... cost of 470 ohm: drive current per gate held at 10 V

PASS ... and across all six driver gates

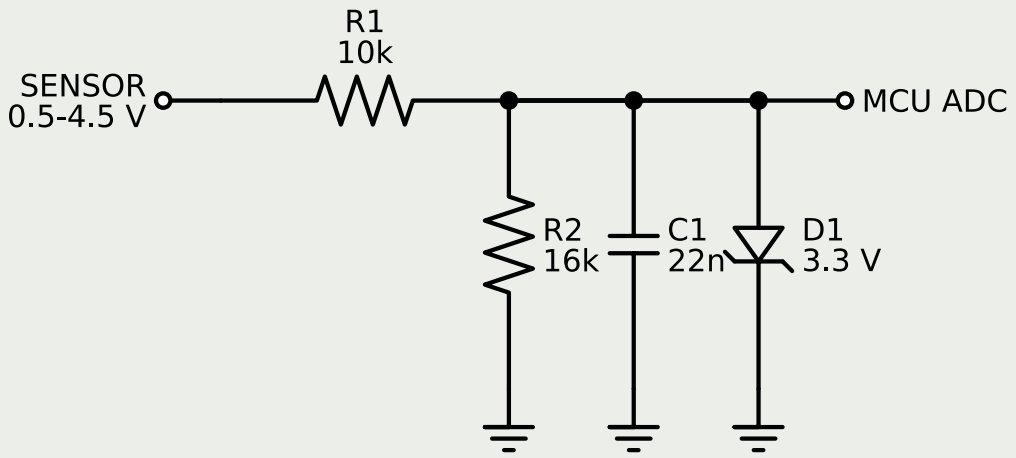
PASS chosen FETs' worst Crss (max) against the 500 pF sizing

Netlist: sim/blocks/supervisor.cir

# 02 Sensor front-ends

Everything that turns something on the engine into a number in the ADC.

## Ratiometric sensor input pins 41, 35, 80, 37 16/16 checks




The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

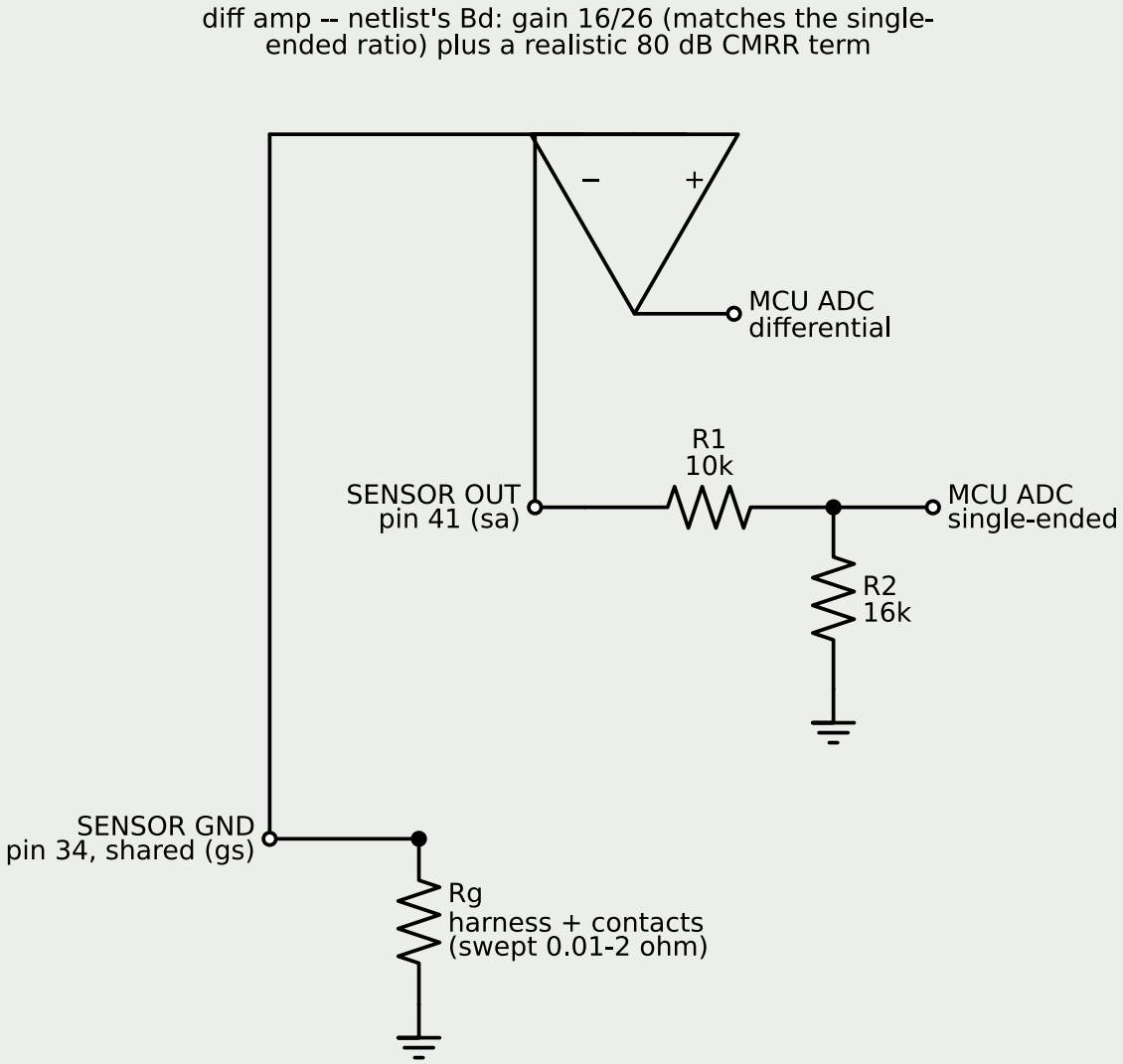
**Why it looks like that.** Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

> **What the model does not cover.** Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

| | |
|---|---|
| PASS sensor zero (0.5 V) maps to | 0.308 V |
| PASS sensor full scale (4.5 V) maps to | 2.769 V |
| PASS ... clears the 3.3 V clamp | 0.531 V |
| PASS span used of ADC range | 74.592 % |
| PASS low-pass corner | 1,174.898 Hz |
| PASS loss at sensor's own 100 Hz | -0.031 dB |
| PASS attenuation at 100 kHz switching | -38.596 dB |
| PASS 40 V harness short: clamp node (one BAV199 above 3.3 V) | 4.326 V |
| PASS ... pad behind 1k | 3.931 V |
| PASS ... current injected into the pad, against +/-3 mA | 0.395 mA |
| PASS ... fault current limited to | 3.567 mA |
| PASS raw reading moves with a +/-5% rail drift | 10.000 % |
| PASS ... so an assumed 5.000 V rail is a 10% error band | 10.0% of reading, and nothing in the raw value reveals it |
| PASS corrected reading is flat | 0.000 % |
| PASS correction recovers the true fraction | 0.500 |
| PASS ... divider ratio cancels, so its tolerance drops out too | measurand = Vsig/Vrail -- k appears in both and divides away |

Netlist: sim/blocks/sensor_ratiometric.cir

## Differential sensor input `pin 34 group` `9/9 checks`




Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

**Why it looks like that.** Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against (+/-2 V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider (R1/R2, the single-ended path) plus a generic amplifier for the differential path,

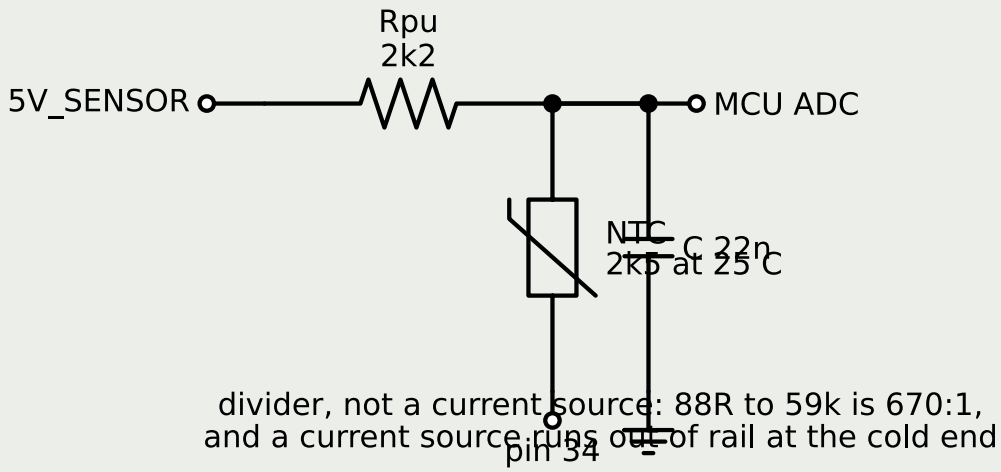
because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

> **What the model does not cover.** The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

| | |
|---|---|
| PASS clean harness (50 mOhm), single-ended error | |
| PASS 1 ohm return, ground offset present | |
| PASS ... single-ended reading error | |
| PASS ... differential reading error, referred to the sensor | |
| PASS ... single-ended, referred to the sensor: the whole offset | |
| PASS differential is better by | |
| PASS ... it was 90x at the inferred 80 dB and 16/26 gain; the chosen part's MINIMUM 88 dB is what moved it | the argument for differential got stronger, not weaker, when a real |
| PASS single-ended error at 2 ohm (bad contact) | |
| PASS ... single-ended needs contact resistance below | ~0.1 ohm forever, to hol |

Netlist: sim/blocks/sensor_differential.cir

## NTC temperature input `pin 79` `11/11 checks`




The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full –40 to 150 C range.

**Why it looks like that.** Memo 09 found DSE drives its temperature senders with a fixed 15 mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer: 88 ohms to 59 kilohms is a 670:1 span against the 50:1 a genset sender covers, and a current source sized to give useful signal at 150 C demands 5.9 V from a 5 V rail at –40 C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

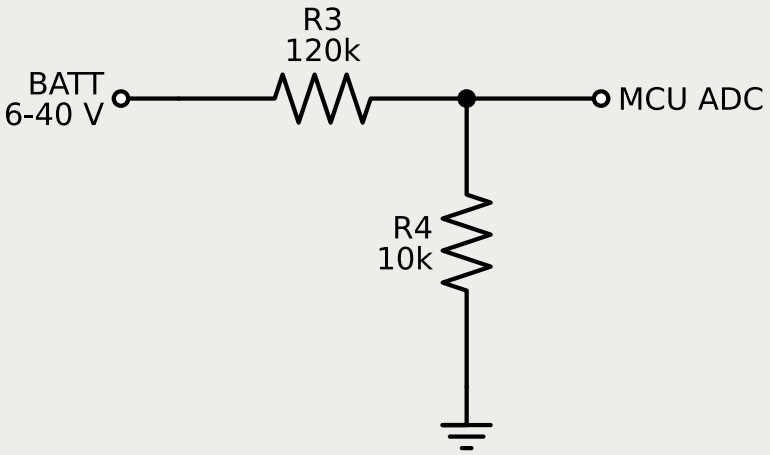
> **What the model does not cover.** Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

| | | |
|---|---|---|
| PASS divider at 150 C (88 ohm) | | 0.192 V |
| PASS divider at 25 C (2.5k) | | 2.664 V |
| PASS divider at -40 C (59k) | | 4.820 V |
| PASS ... never leaves the 5 V rail | | 4.820 V |
| PASS 100 uA source needs this at -40 C | | 5.899 V |
| PASS ... but the rail is 5 V | 5.9 V demanded -- out of compliance, reading clips | |
| PASS 100 uA source at 150 C gives only | | 8.800 mV |
| PASS current that WOULD stay in compliance | | 84.746 uA |
| PASS ... leaving this at 150 C | | 7.458 mV |
| PASS divider self-heating power at 25 C | | 2.829 mW |
| PASS ... error at ~2 mW/C dissipation constant | | 1.414 degC |

Netlist: sim/blocks/ntc_frontend.cir

## Battery voltage sense pins 04, 06

9/9 checks




R3/R4: the adopted divider (spec sec.4, pins 04/06). battery_sense.cir also carries a rejected R1/R2 75k/10k divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

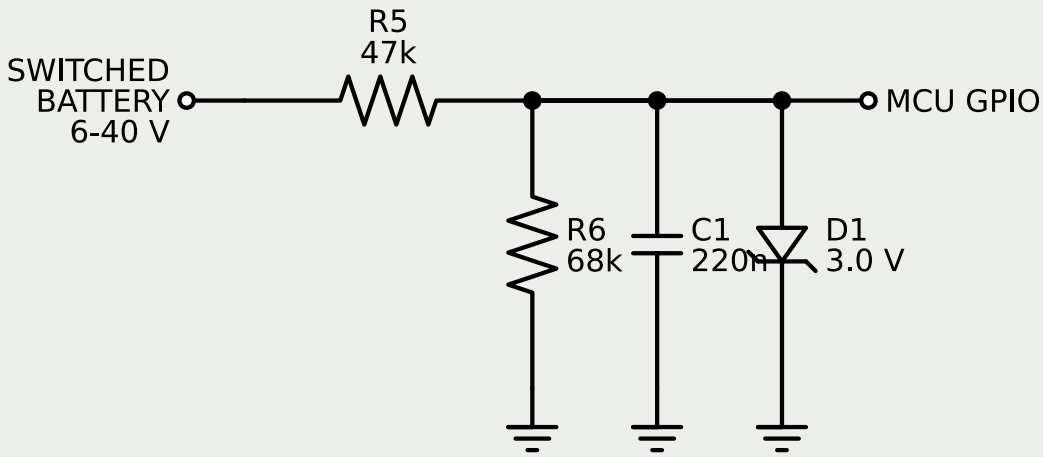
**Why it looks like that.** The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

> **What the model does not cover.** Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

| | | |
|---|---|---|
| PASS | inherited 75k/10k peak at 40 V | 4.694 V |
| PASS | ... and so exceeds a 3.3 V ADC | 4.69 V > 3.3 V -- would damage input |
| PASS | corrected 120k/10k peak at 40 V | 3.069 V |
| PASS | ... stays under the 3.3 V rail | 3.069 |
| PASS | headroom below rail | 0.231 V |
| PASS | reading at 6 V cranking dip | 0.462 V |
| PASS | ... resolves to usable ADC counts | 572.867 LSB |
| PASS | divider is linear (max residual) | 0.000 V |
| PASS | slope matches 10/130 | 0.077 |

Netlist: sim/blocks/battery_sense.cir

## Discrete switch input `pins 20, 24, 71` `15/15 checks`




The switched-battery inputs, active high.

**Why it looks like that.** This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

> **What the model does not cover.** The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

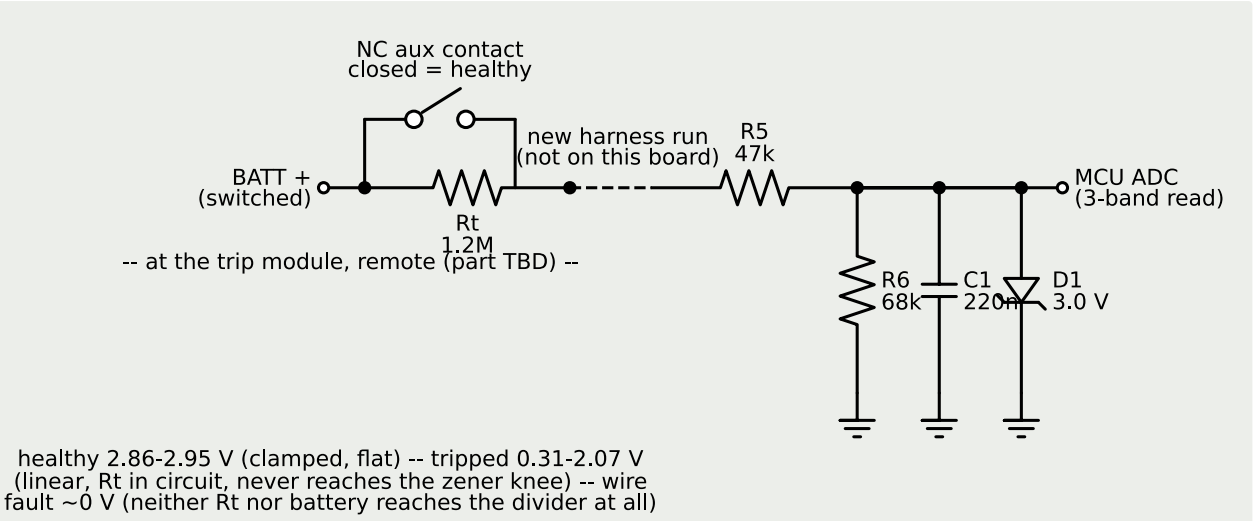
| | | |
|---|---|---|
| PASS | pad max over -24..40 V in (weak, -40 C) | 3.300 V |
| PASS | pad min over -24..40 V in (weak, -40 C) | 0.020 V |
| PASS | pad max over -24..40 V in (typical, 125 C) | 3.300 V |
| PASS | pad min over -24..40 V in (typical, 125 C) | 0.026 V |
| PASS | base at -24 V reverse battery, against VEBO 6 V | -1.047 V |
| PASS | 6 V cranking dip, hFE 30, -40 C: pad reads LOW (active) | 0.027 V |
| PASS | ... input at which the weak cold part reaches VIL | 2.500 V |
| PASS | typical part, 125 C: input that first pulls the pad below VIH | 1.300 V |
| PASS | ... so a harness floating at 1 V still reads inactive | 3.230 V |
| PASS | pull-up channel, contact OPEN, hFE 30, -40 C: pad LOW | 0.055 V |
| PASS | pull-up channel, contact CLOSED, 125 C: pad HIGH | 3.300 V |
| PASS | pad edges through a 5-chatter bounce | 1.000 |
| PASS | settled LOW after the bounce | 0.028 V |
| PASS | debounce time constant (10k \|\| 4.7k) x 4.7 uF | 15.027 ms |
| PASS | contact wetting current at 12 V (≥ 1 mA for non-gold contacts) | 1.120 mA |

Netlist: sim/blocks/discrete_input.cir

## Trip-module sense

new pin, no OEM harness

14/14 checks




A three-state supervised loop reading whether the mandatory standalone overspeed trip module's contacts are open or closed — not a copy of the input above, on purpose.

**Why it looks like that.** Spec §3: U12 closed badly on 18 Sep 2026 — the fitted DSE4522's fuel relay only signals, it does not remove power, so a standalone trip module in series with the fuel metering unit's own battery feed (fuse 8F1) is mandatory again, and this ECU must sense its state rather than read a trip as a fuelling anomaly. discrete_input.cir's own accepted limitation — an open wire reads identically to 'not asserted', accepted there because changing the sense would break OEM harness compatibility — does not carry over: this pin has no OEM harness to stay compatible with, and an undetectable broken wire here would read 'not tripped' at the exact moment a trip occurs, which is the one failure spec §3 exists to catch. So this block adds an end-of-line resistor (Rt, 1.2 MΩ) in parallel with the module's own auxiliary contact, physically at the module end of the new harness run, giving three distinguishable ADC bands on the same R5/R6/zener front end discrete_input.cir already validates: healthy (contact closed, Rt shorted) clamps to 2.86–2.95 V exactly as that file's input does; tripped (contact open, Rt in circuit) is a plain, unclamped linear divider that stays 0.31–2.07 V across 6–40 V, never reaching the zener's knee; wire fault (a break anywhere in the run removes both the contact and Rt the same way) reads ~0 V. A contact-bounce transient, reusing discrete_input.cir's own nasty bounce profile inverted, shows the reading never leaves the healthy band during the bounce itself — C1's

time constant on this branch is long enough to filter chatter shorter than a few hundred microseconds — and settles cleanly into the tripped band only once the contact stays open.

> **What the model does not cover.** No trip module is chosen (memo 02's costed GAC SSW675 / Murphy HD9063 candidates survive costing but neither's aux-contact rating is pulled), so Rt=1.2 MΩ is this file's own INFERRED choice with no datasheet or industry EOL-loop convention behind it. The loop is high impedance (the same few-hundred-microamp class discrete_input.cir's divider runs at), not the low-Z, milliamp-class loop industrial supervised-contact practice normally uses — chosen to reuse the existing front-end rather than build a new one, and the cost is a genuinely tight margin: at the 6 V cranking dip with Rt at +5% tolerance, the tripped band's floor sits only 0.147 V above the wire-fault threshold, about half the nominal-case margin, before any harness leakage is stacked on top. This pin also needs to land on an ADC-capable channel, not a bare GPIO — a pin-mux consideration for whoever assigns the physical PTxx map.

| |
|---|
| PASS healthy (contact closed) band floor |
| PASS healthy band ceiling |
| PASS tripped (Rt in circuit) band floor, at 6V cranking dip |
| PASS tripped band ceiling, at 40V |
| PASS wire fault reads near zero |
| PASS tripped band never reaches the zener's 3.0V knee |
| PASS margin: wire-fault threshold to tripped floor (nominal Rt) |
| PASS margin: tripped ceiling to healthy floor (nominal Rt) |
| |

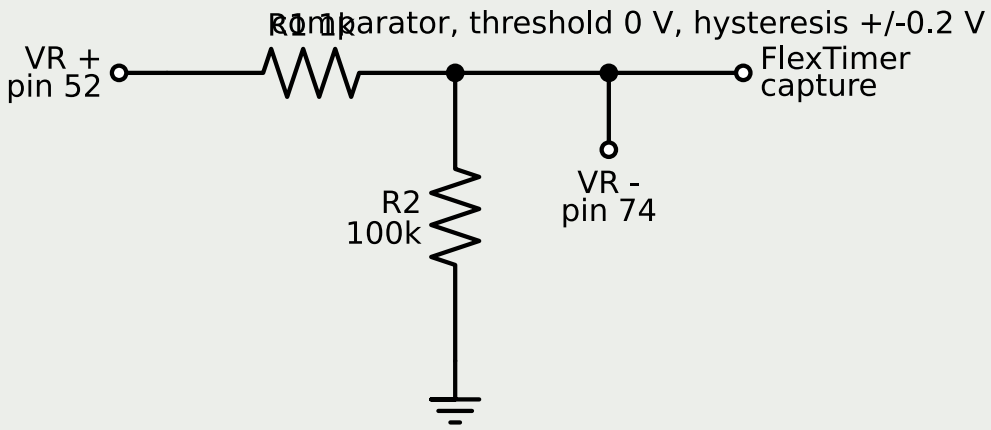
| | |
|---|---|
| PASS ... both bands separated with real margin, not touching | wire-fault margi |
| PASS during bounce (t<4.5ms), reading never leaves the healthy band | |
| PASS settled value (t=100ms) approaches the DC asymptote (hand: 0.6206 V) | |
| PASS ... which sits inside the tripped band, not wire-fault or healthy | |
| PASS ... global minimum over the whole run stays clear of wire-fault | |
| PASS open-wire detection: needed here, unlike pins 20/24/71 | no OEM harness to stay compatible with, and an undetectable open reads |

Netlist: sim/blocks/trip_module_sense.cir

# 03 Speed and position

The most safety-critical input on the board.

## Crank speed conditioner `pins 52, 74, 30` `7/7 checks`




A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

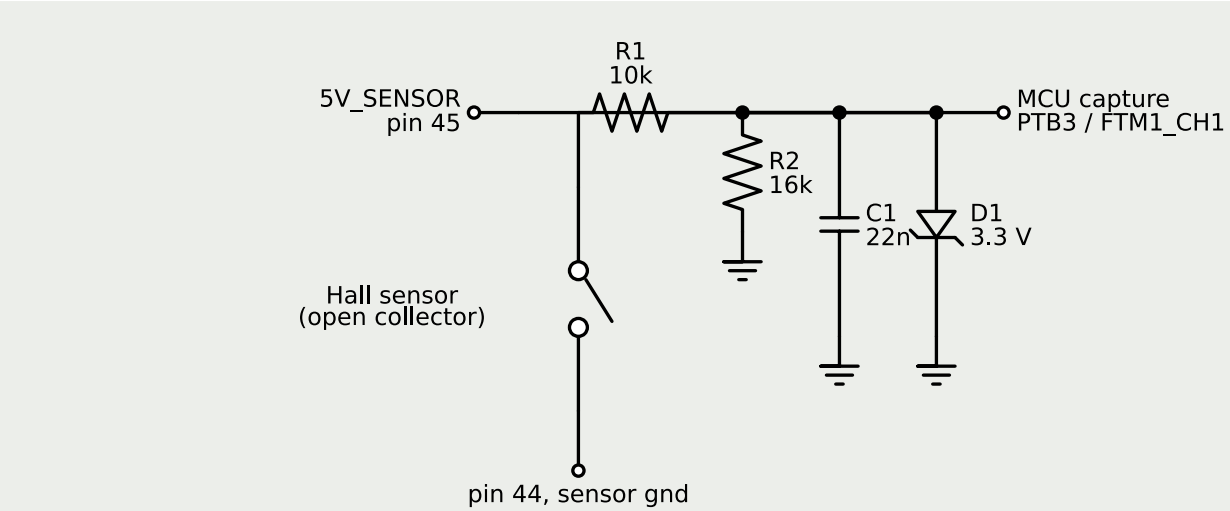
**Why it looks like that.** A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

> **What the model does not cover.** A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

| | | |
|---|---|---|
| PASS zero-cross, running (1500 rpm, 30 teeth) | | 30.000 edges |
| PASS zero-cross, cranking (150 rpm, 3 teeth) | | 3.000 edges |
| PASS fixed 5 V threshold, running | | 30.000 edges |
| PASS fixed 5 V threshold, cranking | | 0.000 edges |
| PASS ... so a fixed threshold never starts the engine | works at 1500 rpm, blind at 150 rpm -- passes on the bench | |
| PASS running tooth period | | 666.667 us |
| PASS ... jitter across the window | | 0.000 us |

Netlist: sim/blocks/vr_conditioner.cir

## Cam signal front-end `pins 45, 46, 44` `12/12 checks`




R1/R2/C1/D1: identical values to sensor_ratiometric.cir's own 5V_SENSOR front end -- same divider ratio, same filter corner, same clamp. Not a coincidence: pin 46 is not 5 V tolerant on this MCU (S32K1xx datasheet Vih max = VDD+0.3 V), so the spec's literal 'pull-up to 5V_SENSOR' has to be divided into the 3.3 V domain before PTB3, the same treatment every other 5 V-rail input gets.

A cam Hall sensor's open-collector output, pulled up and divided into the 3.3 V domain before PTB3/FTM1_CH1, instead of the spec's literal pull-up straight to the 5 V sensor rail.

**Why it looks like that.** PTB3 is not 5 V tolerant for anything this design should rely on — the S32K148's own datasheet puts guaranteed valid-high recognition at VDD+0.3 V, about 3.6 V, and treats a higher DC injection allowance as a fault margin, not a design basis for a pin that sees 5 V every cam revolution for the machine's whole life. Nothing in sim/blocks/ ever modelled this pin before — there is a conditioner for the crank sensor and no cam equivalent — so the overvoltage sat uncaught. The divider (10k/16k) and filter (22 nF, 1176 Hz corner) reuse sensor_ratiometric.cir's own values exactly, since both share the same 5V_SENSOR rail and the same target domain: 94x above the cam's 12.5 Hz signal (half crank speed at 1500 rpm, one lobe per revolution assumed) and still well below switching-frequency noise.

> **What the model does not cover.** The real Hall sensor's saturation voltage, sink current, and internal pull-up (if any) are not modelled — no part chosen. Lobe count is INFERRED at 1 per cam revolution; the TOLERANCE section shows the filter margin survives a 3-lobe alternative just as well. A real cam signal's edge shape and any duty-cycle asymmetry from target

> geometry are not modelled, the same limitation vr_conditioner.cir already states for the crank signal's own non-ideal waveform.

| | |
|---|---|
| PASS sensor off, 5V_SENSOR nominal (5.0 V) → pad | 3.077 V |
| PASS ... clears the 3.6 V abs max | 0.523 V |
| PASS sensor_rail.cir's own fault floor (4.5 V) → pad | 2.769 V |
| PASS ... still reads logic high (⩾2.31 V, 0.7*VDD) | 2.769 V |
| PASS above-nominal rail (5.25 V) → pad | 3.231 V |
| PASS ... still clears the 3.6 V abs max | 0.369 V |
| PASS ... WAS: the zener's loading of the signal at this point | 0.084 V |
| PASS never exceeds 3.6 V abs max in transient | 3.077 V |
| PASS rising edges, 170 ms window at 12.5 Hz (80 ms period) | 2.000 edges |
| PASS 40 V harness short: clamp node (BAV199-Q to 3.3 V) | 4.326 V |
| PASS ... current injected into the pad behind 1k, vs +/-3 mA | 0.395 mA |
| PASS ... fault current limited to | 3.567 mA |

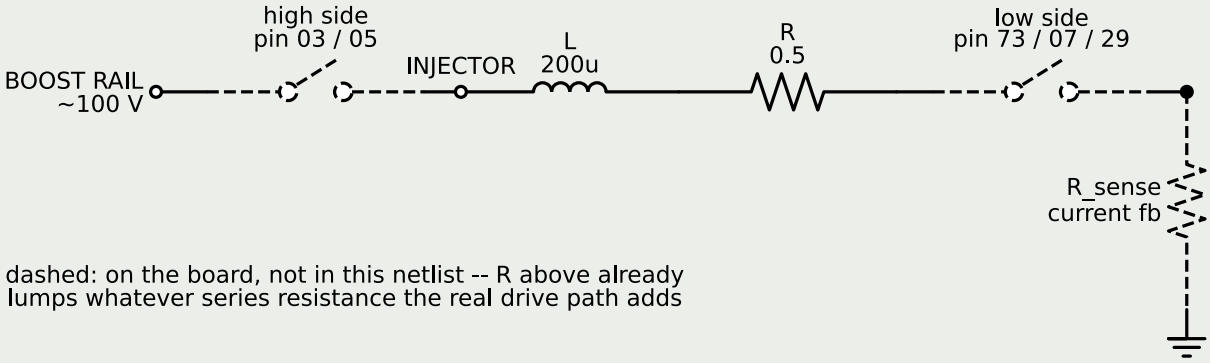
Netlist: sim/blocks/cam_frontend.cir

# 04 Injection

The most expensive stage on the board and the reason it exists.

## Why the boost rail exists `pins 03, 05`

13/13 checks




The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

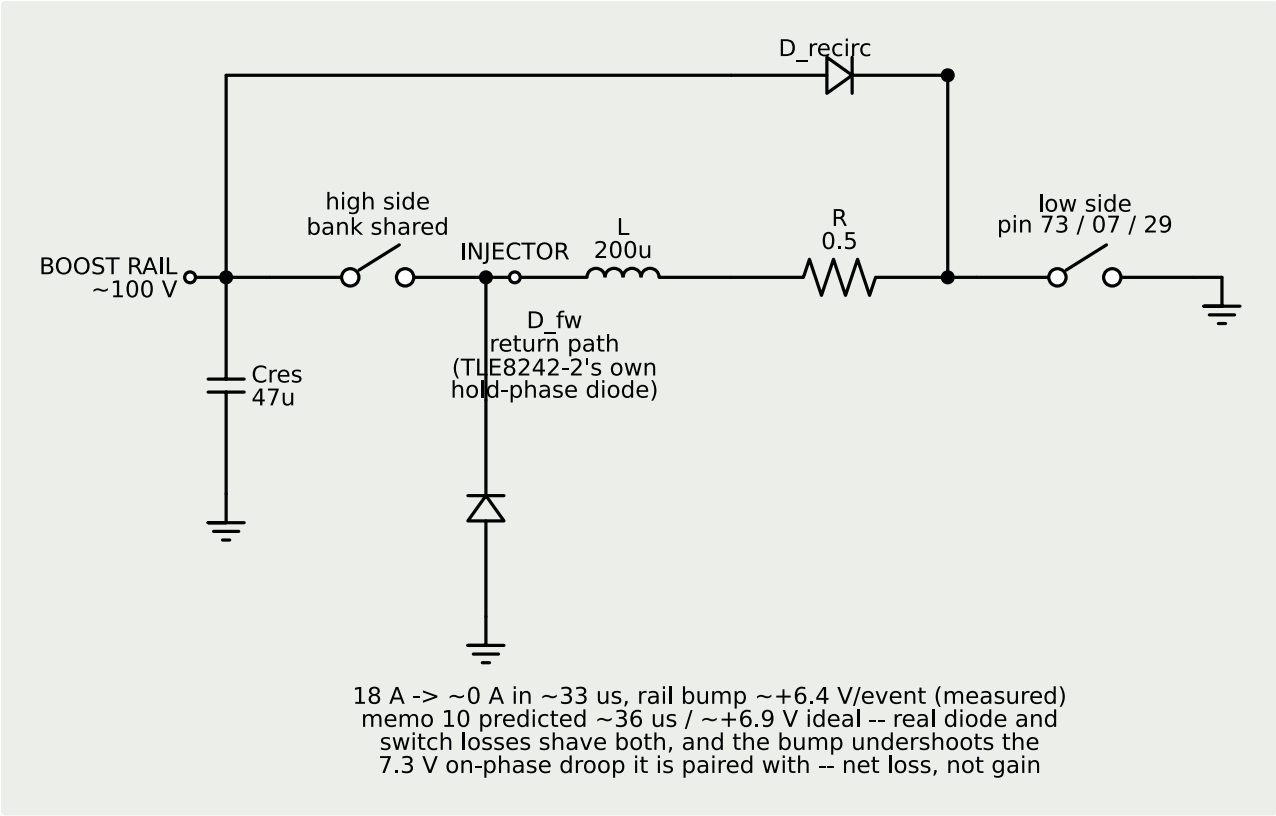
**Why it looks like that.** 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

> **What the model does not cover.** An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

| | |
|---|---|
| PASS time to 18 A peak from 13.5 V battery | 439.496 us |
| PASS time to 18 A peak from 100 V boost | 37.796 us |
| PASS boost is faster by | 11.628 x |
| PASS battery drive eats this much of a 1 ms injection | 43.950 % |
| PASS boost drive eats this much of a 1 ms injection | 3.780 % |
| PASS ... evidence: ratio at -40C ambient, vs the 11.6x NOMINAL (not a requirement) | 9.856 x |
| PASS ... REQUIREMENT: boost still decisively faster at -40C (>5x) | 9.856 x |
| PASS ... and battery drive still eats this much of a 1 ms injection at -40C | 36.760 % |
| PASS ... time to 18 A from 100 V boost, -40C ambient | 37.296 us |
| PASS ... evidence: ratio at +125C ambient, vs the 11.6x NOMINAL (not a requirement) | 19.017 x |
| PASS ... REQUIREMENT: boost still decisively faster at +125C (>5x) | 19.017 x |
| PASS ... and battery drive still eats this much of a 1 ms injection at +125C | 73.210 % |
| PASS ... time to 18 A from 100 V boost, +125C ambient | 38.496 us |

Netlist: sim/blocks/injector_boost.cir

## Turn-off / recirculation `pins 73, 07, 29` `15/15 checks`




The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47 uF) instead of a clamp that just burns it — plus a second diode (D_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

**Why it looks like that.** Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-“Questionable”) and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33 us to collapse 18 A to zero (memo predicted ~36 us), 25x faster than freewheeling to the battery (830 us — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost_converter.cir's charge regulation (Bchga) has no bleed path either way; it just happens not to need one for the peak-only case.

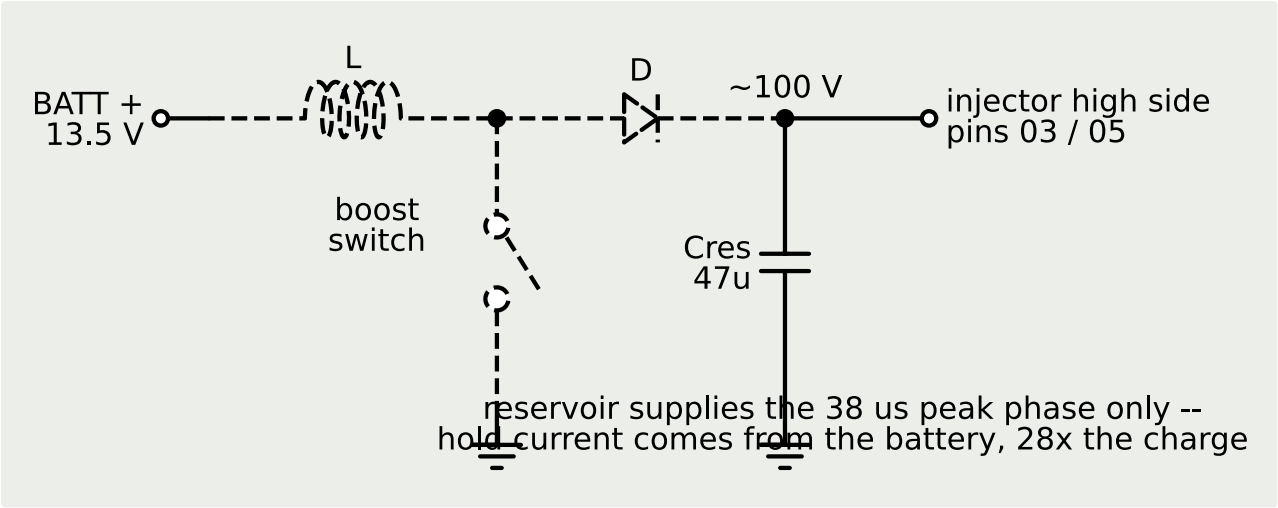
> **What the model does not cover.** The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any dV/dt read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery (Qrr), the diode's thermal transient, and — inherited from injector_boost.cir — magnetic saturation and needle motion.

| | |
|---|---|
| PASS turn-off time, 18 A → ~0 A on the boost rail | |
| PASS ... vs freewheel-to-battery baseline (comparison only) | |
| PASS ... boost recirculation is faster by | |
| PASS on-phase droop, pilot event | |
| PASS ... matches boost_converter.cir's own droop figure | |
| PASS recovery bump, pilot event (isolated) | |
| PASS ... undershoots the droop it is paired with (real losses) | |
| PASS rail never exceeds 100 V setpoint (3-event circuit) | |
| PASS rail recovered by the next cylinder (26.67 ms after post) | |
| PASS hold-current cutoff into the SAME cap, no offsetting draw | |
| PASS ... THIS is the scenario that overshoots 100 V | |
| PASS ... so the verdict hinges on whether hold-current cutoffs share the boost-recirculation diode | peak-only recirculation never overshoots; hold-current reci |
| PASS peak instantaneous power in D_recirc during turn-off | |
| PASS ... exceeds the ~3.6 W worst-case AVERAGE by | |

```
PASS differential-
mode edge vs                                  not modelled here (ideal switches) --
emi_filter.cir
```

Netlist: sim/blocks/injector_turnoff.cir

## Boost rail reservoir `pins 03, 05` `95/95 checks`




The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

**Why it looks like that.** The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

> **What the model does not cover.** Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

| |
|---|
| PASS droop, peak phase only (47 uF) |
| PASS ... rail stays usefully above battery |
| PASS droop, peak AND hold from the rail |
| PASS ... so the rail collapses |
| PASS hold phase costs this much more charge |
| PASS recovered by the next injection (26.67 ms) |
| PASS ... 50 mA average charging is enough |
| PASS one hold-cutoff event puts this on the rail |
| PASS without a bleed, three events stack to |
| PASS ... and it STAYS there -- nothing can discharge it |
| PASS with a 22k bleed, same three events reach |
| PASS ... but the rail is back at setpoint before the next cylinder |
| |

| | | |
|---|---|---|
| PASS time to clear one event | | |
| PASS ... against the 240-degree budget at 1500 rpm | | |
| PASS ... cost, burned continuously whenever the rail is up | | |
| PASS TPS40210-Q1 VDD absolute maximum | | |
| PASS ... against the pulse 2a rail it would otherwise sit on | | |
| PASS ... so it CANNOT go straight on VBAT_PROT | same shape as DRV8873-Q1's 40 V VM. | |
| PASS WAS: 47 ohm + BZG03C43-M, VDD at pulse 2a, hot | | |
| PASS ... over the 52 V absolute maximum by | | |
| PASS ... and its 40 V minimum conducts on the load dump it was chosen to stay out of | | 43 |

PASS VDD_BAT
at pulse 2a:
180 ohm +
BZG03C36, VZ
max, 125 C,
ZZ bound
(reaches VDD
through the
Schottky)

PASS ...
inside the
absolute
maximum by

PASS ...
zener current
during the 50
us pulse

PASS ...
series
resistor,
peak
dissipation
for 50 us

PASS load
dump, VZ min
at -40 C:
zener current
for 400 ms

PASS ...
zener
dissipation
for 400 ms
(1.25 W is
the STEADY
rating at 25
C; this leans
on thermal
mass, not
read off a
Zth curve)

PASS start-up
VDD at the 6
V cranking
dip (Qg typ,
through the
Schottky)

PASS ...
above the
UVLO turn-on
ceiling by

```
PASS ... at
twice the
typical gate
charge
(DMN15H310SK3
gives no
maximum) --
THIN, stated
```

```
PASS duty
needed at the
6 V cranking
corner
```

```
PASS ... at
13.5 V, for
contrast
```

```
PASS duty the
part allows
at 150 kHz
```

```
PASS ...
margin at the
cranking
corner
```

```
PASS ... so
the cranking
corner is
reachable
```

```
PASS WAS: 330
uH, VDD 5.16
V on the
battery: 100%
RISNS(max) at
6 V
```

```
PASS ... the
drawn 82 mOhm
exceeded it
-- subharmonic
while
cranking
```

```
PASS VDD from
the output
follower,
lowest (VZ
min, -40 C,
low current,
cold Vbe, 56
ohm drop)
```

```
PASS ...
highest (VZ
max, 125 C,
hot Vbe)
```

PASS ... inside the 52 V absolute maximum by

PASS ... still regulating at the rail's worst droop (92.7 V), headroom

PASS ... pass transistor, normal (104 V rail, lowest VDD)

PASS ... junction at a 105 C board (104 K/W)

PASS ... VDD shorted: limited to one Vbe across 56 ohm

PASS ... pass transistor into the short

PASS ... each 3.3k 2512 into the short (0.59 W allowed at 105 C)

PASS IS: 80% RISNS(max), worst corner (VDD 34.3 V, 120 uH, 120 kHz, 6 V)

PASS ... 62 mOhm inside it by

PASS Eq. 49: peak inductor current at the cranking corner

```
PASS ...
sense ceiling
so the limit
never trips
in normal
running
```

```
PASS ... 62
mOhm inside
it by
```

```
PASS current
limit,
highest (180
mV)
```

```
PASS ... past
L401's 10%
saturation
(2.7 A typ)
by -- stated
in decisions
log 8
```

```
PASS ...
under its 30%
saturation
(3.3 A typ)
by
```

```
PASS ...
start-up is
the
exception:
until the
output passes                                          subharmonic is possible for
about 36 V,
VDD is the
battery feed
and the ramp
is shallow
```

```
PASS Q401
gate charge
(typ) under
the 25 nC
ceiling by
```

```
PASS input
current at
the cranking
corner (50 mA
out, 85%)
```

```
PASS duty
needed with
switch, sense
and L401
drops, hot
```

| | |
|---|---|
| PASS ... inside what the part allows at the +7% frequency corner by | |
| PASS the same 200 ns off-time at 2.2 MHz allows only | |
| PASS ... which does not reach even the 13.5 V duty | |
| PASS ... so switching SLOWER is the requirement here | 200 ns of off-time is a floor in time, no |
| PASS bank node worst case (reference +2%, one recirculation bump) | |
| PASS ... so VB reaches | |
| PASS AUIRS2181S: VB absolute maximum 625 V, margin | |
| PASS UCC27211A-Q1, the obvious 120 V candidate: HB margin | |
| PASS ... 4 V of headroom on an ABSOLUTE maximum -- rejected | |
| PASS bank node's lowest point, D_fw conducting 18 A | |

| | |
|---|---|
| PASS ... AUIRS2181S logic operational down to -5 V; clear by | |
| PASS ... UCC27211A-Q1's HS DC minimum is -1 V -- violated in NORMAL operation, not in a fault | |
| PASS cross-conduction interlock on AUIRS2181S | none (datasheet feature table: 2181 'no', 2183/2184 'yes') -- UCC27 |
| PASS bootstrap droop over the 38 us peak phase (SQM85N15-19 120 nC max, 150 uA IQBS, 1 uF) | |
| PASS ... as a fraction of the ~11.5 V bootstrap supply | |
| PASS why not the LM5164 already on the board: its VIN abs max | |
| PASS ... against the rail it would sit on | |
| PASS 12V_GATE above the driver's UVLO+ ceiling (9.8 V) by | |
| | |

```
PASS 12V_GATE
low corner
(BZX84-C13-Q
min, low
current, hot
Vbe)
PASS ...
above the
driver UVLO+
ceiling (9.8
V) by
PASS 12V_GATE
high corner
(C13-Q max,
cold Vbe)
PASS ...
under the
chosen FETs'
20 V Vgs
rating by
PASS current
limit: one
Vbe across 56
ohm
PASS ...
against the 5
mA design
load
PASS WAS:
pass
transistor
alone, normal
(88 V at 5
mA)
PASS ... over
what PZTA42-Q
may dissipate
at a 105 C
board
PASS ... and
shorted, held
by the limit
PASS IS:
2x3.3k
collector
resistors --
transistor,
normal
PASS ...
junction at a
105 C board
```

```
PASS ...
transistor
into a
shorted
output
```

```
PASS ...
junction,
shorted, 105
C board
```

```
PASS ... each
2512
resistor,
shorted (0.59
W allowed at
105 C on a 1
W part
derated
linearly 70-
155 C)
```

```
PASS ...
lowest boost
rail that
still
regulates
12V_GATE
```

```
PASS ...
against the
rail's worst
droop (100 -
7.3 V)
```

```
PASS standing
load on the
boost: bleed
+ gate rail +
divider + its
own VDD
follower and
zener bias
```

```
PASS ... so
the peak-
phase draw
now recovers
in
```

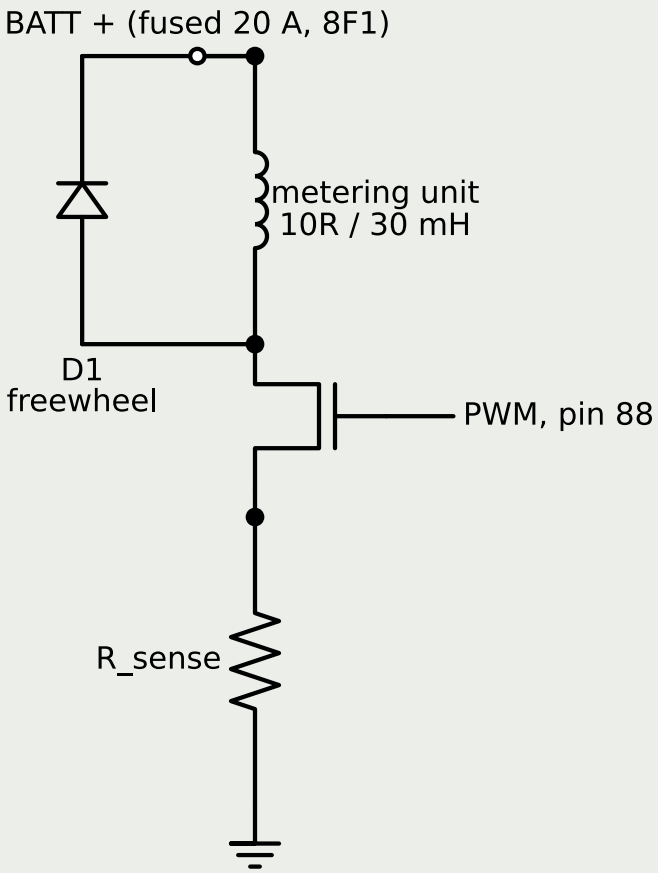
```
PASS ...
still inside
the 26.67 ms
between
cylinders by
```

Netlist: sim/blocks/boost_converter.cir

# 05 Actuators

Outputs that move something.

## Fuel metering unit driver `pin 88` `25/25 checks`




Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

**Why it looks like that.** Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

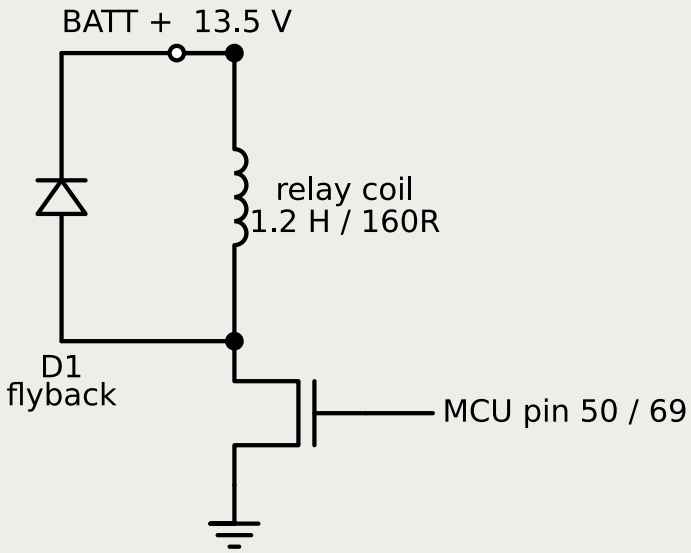
> **What the model does not cover.** The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

> rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

| | | |
|---|---|---|
| PASS mean current at 100 Hz | | 0.642 A |
| PASS mean current at 1 kHz | | 0.637 A |
| PASS mean current at 10 kHz | | 0.635 A |
| PASS ripple at 100 Hz | | 0.969 A |
| PASS ... as fraction of mean | | 150.919 % |
| PASS ripple at 1 kHz | | 0.118 A |
| PASS ripple at 10 kHz | | 0.012 A |
| PASS ripple scales as 1/f | | 9.965 x |
| PASS 100 Hz is not dither, it is chopping | 151% ripple -- valve follows the PWM | |
| PASS ... evidence, OPEN LOOP: mean current at 100 Hz, -40C ambient | | 0.854 A |
| PASS ... evidence, OPEN LOOP: mean current at 1 kHz, -40C ambient | | 0.862 A |
| PASS ... evidence, OPEN LOOP: mean current at 10 kHz, -40C ambient | | 0.858 A |
| PASS ... so open-loop duty misses the 0.675 A setpoint at -40C by | | 27.132 % |
| PASS ... control law: frequencies still agree, -40C | | 0.992 % |
| PASS ... RESOLVED, CLOSED LOOP: mean current at -40C | | 0.679 A |
| PASS ... evidence, OPEN LOOP: mean current at 100 Hz, +125C ambient | | 0.458 A |
| PASS ... evidence, OPEN LOOP: mean current at 1 kHz, +125C ambient | | 0.462 A |
| PASS ... evidence, OPEN LOOP: mean current at 10 kHz, +125C ambient | | 0.459 A |
| PASS ... so open-loop duty misses the 0.675 A setpoint at +125C by | | -31.883 % |
| PASS ... control law: frequencies still agree, +125C | | 0.727 % |
| PASS ... RESOLVED, CLOSED LOOP: mean current at +125C | | 0.682 A |
| PASS closed loop needs duty headroom: max coil current at +125C | | 0.977 A |
| PASS ... setpoint as a fraction of that ceiling | | 69.100 % |
| PASS closed loop at nominal 25C | | 0.679 A |
| PASS ... hysteresis band it actually regulates within | | 0.133 A |

Netlist: sim/blocks/metering_unit_pwm.cir

## Relay driver `pins 50, 69` `6/6 checks`




A low-side FET switching a relay coil, run with and without the flyback diode.

**Why it looks like that.** With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

> **What the model does not cover.** The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

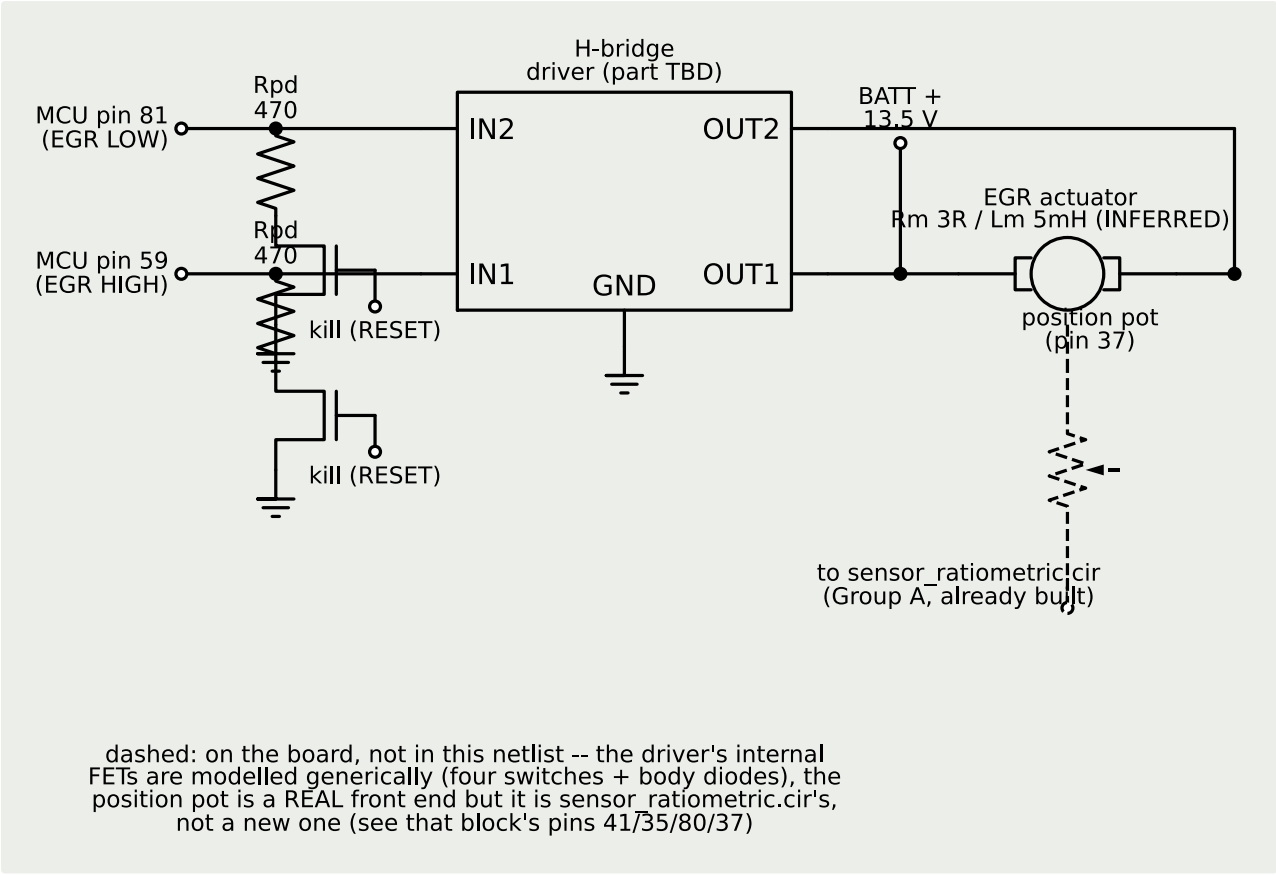
| | |
|---|---|
| PASS drain peak WITH flyback diode | 14.274 V |
| PASS drain peak WITHOUT diode | 402,899.522 V |
| PASS ... diode suppresses the kick by | 28,226.510 x |
| PASS gate overdrive from a bare 3.3 V GPIO (was) | 0.600 V |
| PASS gate overdrive from the AHCT buffer at VCC 4.5 V (is) | 1.700 V |
| PASS ... FET drop at coil current, Rds(on) at the 4.5 V point | 36.450 mV |

Netlist: `sim/blocks/relay_driver.cir`

## EGR actuator H-bridge

pins 59, 81, position feedback pin 37

19/19 checks




Two logic/PWM lines into an external H-bridge driving a positional DC-motor actuator both ways — the one output on this board that is not a low-side switch — modelled as four generic switch positions with anti-parallel body diodes, plus the same active kill-clamp (Rpd 470 Ω, keyed to the supervisor's RESET) spec §4 requires at every driver gate.

**Why it looks like that.** This is a positional actuator, not an on/off solenoid: it is held at a commanded position, which is why it needs a bridge. Spec §4's outputs table lists pins 59/81 in the same row set as the six gates supervisor.cir already protects, but that file's own header enumerates only 88/73/07/29/03/05 — EGR was never added because this block did not exist. Reusing supervisor supervisor.cir's validated kill topology closes that gap rather than re-deriving it. The two-input "sign" scheme this design assumes (IN1 drives the S1+S4 diagonal, IN2 drives S3+S2) means no combination the ECU can present ever commands one leg's own high and low side together — real shoot-through is a property of the eventual driver IC's internal interlock, a part not yet chosen, not something these two lines can cause on their own. Stall current (jammed, or simply sitting at its own end of travel — normal operation for a positional actuator, not a fault) comes out to 4.36 A at an assumed 3 Ω armature resistance, comfortably past a class-typical 2–3 A small-

actuator driver rating: the eventual part needs active current limiting, not a fixed series resistor. Commanding coast with current already flowing is checked with and without the body diodes, the same comparison relay_driver.cir runs for its flyback diode: with them the bridge nodes stay within a diode drop of the rail; without them the ideal-switch model produces a multi-megavolt artifact, because the motor's stored energy has nowhere else to go.

> **What the model does not cover.** No H-bridge driver IC or FET is chosen — Ron, the body-diode figures, and the motor's Rm/Lm are all INFERRED class figures, the same gap as every other unselected FET on this board. Whether the eventual driver is integrated or discrete FETs behind a gate driver is undecided (memo 12 settled this for the injector and metering stages but not for EGR), so the 470 Ω kill-clamp value here is a uniform-standard choice, not a fresh Miller-coupling derivation for the 13.5 V domain EGR actually runs in. No mechanical model of the actuator exists — no return-spring constant, no gear ratio, no end-stop switches — so "coast lets the spring close the valve" is stated as the assumed fail-safe direction, not simulated. Position feedback on pin 37 is not modelled in this file at all: it is Group A ratiometric, already built and checked in sensor_ratiometric.cir, and building a second front end here would duplicate a proven circuit rather than reuse it.

| | | |
|---|---|---|
| PASS kill-clamp: gate below Vgsth (hand estimate 22.8 ns) | | 22. |
| PASS ... adversarial driver (100R/10V) still held below Vgsth | | 0 |
| PASS ... adversarial settle voltage (hand: 10*4.95/104.95=0.471 V) | | 0 |
| PASS stall current (Vemf=0, hand: 13.5/3.1) | | 4 |
| PASS running current (Vemf=9V, hand: 4.5/3.1) | | 1 |
| PASS coast current (IN1=IN2=0) | | 0 |
| PASS stall current exceeds a class-typical 2-3 A driver rating | | 4 |
| PASS ... so the eventual driver needs active current limiting, not a fixed series resistor | 4.35 A at Rm=3 ohm alone exceeds a 2-3 A class | |
| PASS forward steady current at t=19.9ms matches section 2's stall figure | | 4 |
| PASS WITH diodes: low side stays within a diode drop of ground (hand: -1.09 V) | | -1 |
| PASS WITH diodes: high side stays within a diode drop of the rail (hand: 14.59 V) | | 14 |
| PASS WITHOUT diodes: low side swings far past ground | | 9,038,255 |
| PASS WITHOUT diodes: high side swings far past the rail | | 9,038,269 |
| PASS ... recirculation diodes are load-bearing, not decorative | bounded to 14.6 V with diodes vs 9.04e+06 V w | |

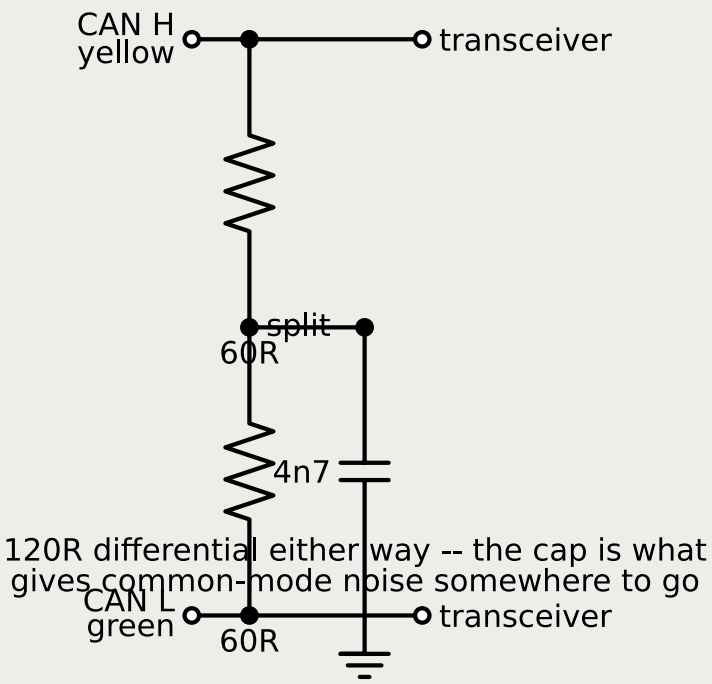
| | |
|---|---|
| PASS position feedback (pin 37) is Group A ratiometric, proven in sensor_ratiometric.cir | 0.308-2.769 V into the 3.3 V ADC, see check_sensor_ratiome |
| PASS rejected naive wiring: rail current with IN1=IN2 asserted | 270 |
| PASS ... matching the two-legs-of-2x50mOhm hand figure | 270 |
| PASS ADOPTED: decoded DIR/PWM driver, both inputs asserted | 4 |
| PASS ... so decode removes the fault state entirely | 270 A naive vs 4.35 A decoded |

Netlist: sim/blocks/egr_hbridge.cir

# 06 Communications

The bus to the genset controller.

## CAN split termination `CAN H, CAN L` 11/11 checks




A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

**Why it looks like that.** Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

> **What the model does not cover.** Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

| | |
|---|---|
| PASS split termination, differential at 250 kbit/s | 119.993 ohm |
| PASS single 120R, differential at 250 kbit/s | 119.993 ohm |
| PASS ... identical differentially | 0.000 ohm |
| PASS split termination, common-mode at 10 MHz | 30.189 ohm |
| PASS single 120R, common-mode at 10 MHz | 500.000 kohm |
| PASS ... split shunts common mode better by | 16,562.490 x |
| PASS split common-mode at the bit rate stays high | 138.099 ohm |
| PASS ... evidence: at 200 ppm/C thick-film tempco (matched direction, -40/125C) -- the corner that forced the requirement | 123.952 ohm |
| PASS ... so ordinary thick-film misses J1939's own 120 ohm by | 3.952 ohm |
| PASS RESOLVED: specified 0.1%/50 ppm thin-film, worst-case stacked | 121.104 ohm |
| PASS ... margin remaining inside the 120 +/-2 ohm window | 0.896 ohm |

Netlist: sim/blocks/can_termination.cir